

ARRAYS

Q9) Boolean Matrix Row Column Modification

DS	Session	Arrays	Question Information	Level 1	Challenge 29
Problem Description: Tina has given a boolean matrix <code>mat[p][q]</code> of size <code>P X Q</code> to Laasya, She wants to modify it such that if a matrix cell <code>mat[m][n]</code> is 1 (or true) then make all the cells of <code>m_th</code> row and <code>n_th</code> column as 1. Can you help Laasya? Constraints: <code>1 <= p, q <= 1000</code> Input Format: First line of the input is the how many rows and columns in Laasya matrix. After that, each line will represent each row and Laasya need to enter numbers. Output Format: Print the resultant matrix output in a separate lines.					
Logical Test Cases Test Case 1 INPUT (STDIN) 2 2 1 0 0 1 EXPECTED OUTPUT 1 1 1 1 Test Case 2 INPUT (STDIN) 3 3 0 0 0 0 1 0 0 0 1 EXPECTED OUTPUT 0 1 1 1 1 1 1 1 1					
Mandatory Test Cases Test Case 1 KEYWORD int arr[r][c]; Test Case 2 KEYWORD arrTemp[r][c] Test Case 3 KEYWORD for(m=0;m<r;m++)					

CODE:

```
C main.c
1  #include <stdio.h>
2
3  int main(void) {
4      int r, c;
5      if (scanf("%d %d", &r, &c) != 2) return 0;
6
7      int arr[r][c];
8      int arrTemp[r][c];
9
10     int m, n;
11
12     for(m=0;m<r;m++) {
13         for (n = 0; n < c; n++) {
14             scanf("%d", &arr[m][n]);
15             arrTemp[m][n] = 0;
16         }
17     }
18
19     for (m = 0; m < r; m++) {
20         for (n = 0; n < c; n++) {
21             if (arr[m][n] == 1) {
22                 for (int j = 0; j < c; j++) {
23                     arrTemp[m][j] = 1;
24                 }
25                 for (int i = 0; i < r; i++) {
26                     arrTemp[i][n] = 1;
27                 }
28             }
29         }
30     }
31
32     for (m = 0; m < r; m++) {
33         for (n = 0; n < c; n++) {
34             printf("%d ", arrTemp[m][n]);
35         }
36         printf("\n");
37     }
38
39     return 0;
40 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
3 3
0 0 0
0 1 0
0 0 1
0 1 1
1 1 1
1 1 1
```